Speed-Oriented Lightweight Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— The lightweight model for salient object detection in optical remote sensing images (SOD-RSI) is a recent emerging topic. Due to the complexity of the task, recently published works have achieved effective model compression but have not yet achieved the desired detection speed. To truly release the detection speed of lightweight models while ensuring a favorable accuracy-efficiency tradeoff, we propose a new speed-oriented lightweight SOD-RSI network (SOLNet), which has significant advantages in detection speed. Specifically, we design a lightweight group attention (LGA) module to deconstruct–interact–recombine channel features and an enhanced dynamic encoding (EDE) module for dynamically capturing spatial information. On this basis, the dynamically enhanced aggregation module (DEAM) is further proposed, which mines the intrinsic correlation of feature information by decoding high-level feature maps, eliminating the need to pay additional attention to other scales. SOLNet completes lightweight and efficient decoding through simple cascade aggregation operations. Notably, we also propose an evaluation strategy that takes both speed and accuracy into account, extending a novel lightweight gain (Lg) metric for SOD-RSI. This not only effectively reveals the under-gain issue of lightweight models but also provides theoretical support for the evaluation of subsequent lightweight works. Experimental results on the challenging EORSSD and ORSSD datasets show that SOLNet achieves significant speed improvements and is the state-of-the-art (SOTA) lightweight SOD-RSI method. The code is available at https://github.com/SpiritAshes/SOLNet.**

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I. Introduction

SALIENT object detection (SOD) is a task that imitates the biological characteristics of human eyes to capture information about objects of interest in the scene [1], [2], [3]. Due to the efficiency and simplicity of the SOD gaze attention mechanism, it has been successfully applied in many computer vision detection fields, such as object tracking [4], [5], [6], image segmentation [7], [8], [9], image quality assessment [10], [11], [12], and image retargeting [13], [14].

Recently, with advancements in technology, several works [15], [16], [17], [18], [19], [20], [21], [22] have provided a solid foundation for SOD in natural scene images (NSIs). However, the fundamental differences between natural and remote sensing images (RSIs) present significant challenges for SOD methods originally designed for NSIs. First, in RSIs, salient objects such as vehicles, ships, and buildings are usually small and sparsely distributed across large scenes, occupying only a small portion of the image. This makes it challenging for SOD-RSI models to focus on the most relevant regions. Second, the high visual similarity between different object categories, such as ships and docks or roads and rivers, makes it difficult for models designed for NSIs to achieve the fine-grained distinctions required for RSIs. Third, compared to NSIs with clear foreground–background separation, the complex background of RSIs further increases the difficulty of segmenting salient objects from the surrounding environment.

Consequently, SOD models specifically designed for optical RSIs have received more and more attention [23], [24], [25], [26], [27].

Currently, building detection models based on convolutional neural networks (CNN) is the mainstream trend in implementing SOD-RSI. CNN models have advantages in modeling the dependencies of local features in images. For example, the end-to-end CNN model LV-Net [28] enhances the local features of the image by designing a bidirectional pyramid module and a nested connected detection module to perceive objects of interest at different scales. However, merely modeling local features is insufficient for effective detection. Therefore, most CNN models design different types of attention mechanisms to capture the long-distance dependencies of features, for example, dense attention fluid

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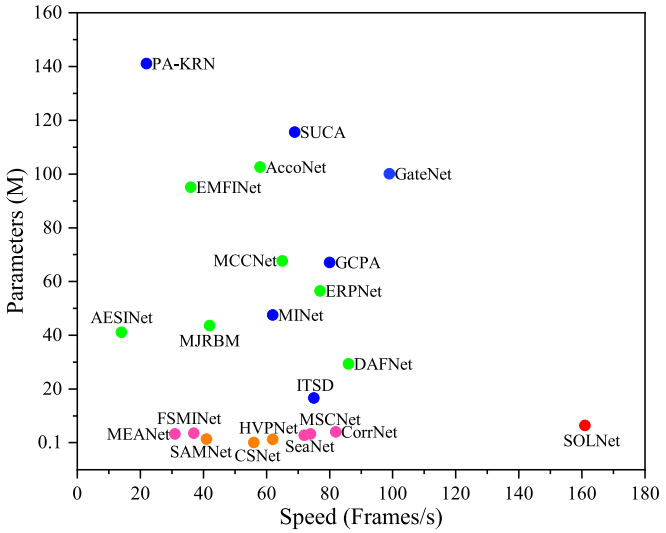


Fig. 1. Performance comparison of the proposed method with related SOD methods. The same color represents the same type of method, and the proposed method is individually represented in red.

network (DAFNet) [29], multiscale context network (MSC-Net) [30], and recent work adjacent context coordination network (ACCoNet) [31]. Moreover, given the impressive capabilities of the vision transformer (ViT) architecture in modeling global information across various vision tasks, recent works [32], [33] have explored the application of self-attention mechanisms based on ViT in SOD-RSIs. ViT models excel at capturing long-range dependencies and global context but may struggle to effectively capture fine-grained local features, particularly in sparse and imbalanced remote sensing data. As a result, they are typically employed in conjunction with CNN architectures in hybrid models, leveraging the strengths of both to enhance global and local feature extraction, thereby improving detection performance.

Notably, the aforementioned works have primarily focused on improving the detection accuracy of SOD-RSIs, but they have paid limited attention to model size and detection speed. This limits the potential scalability of these models in future applications. For lightweight SOD models, the hierarchical visual perception network (HVPNet) [34], originally developed for SOD-NSIs, struggles to adapt well to the SOD-RSI task due to fundamental differences in data characteristics. In the field of SOD-RSIs, the correlation network (CorrNet) [35] provides a noteworthy solution by reducing model complexity through the use of a lightweight subnet dedicated to feature extraction. Building on this, SeaNet [1], which incorporates semantic matching and edge alignment, further reduces network size. Interestingly, while substantial improvements in detection speed would be expected, an exception occurs. As shown in Fig. 1, the uncompressed model multicontent complementation network (MCCNet) [36], edge-guided recurrent positioning network (ERPNet) [25], and DAFNet [29] achieve comparable performance to SeaNet [1] and CorrNet [35] in terms of detection speed. The existence of the above special cases motivates us to further explore lightweight improvement, as our goal is to develop a method that not only optimizes accuracy but also genuinely improves speed.

Specifically, our motivation stems from two key aspects. First, the current improvements in lightweight methods are overly conservative. Although computational parameters have been reduced through sparse computation, the actual detection speed has not significantly improved due to the re-encoding of multilevel features. Imagine if optimization focused on reducing the number of decoding steps and simplifying the decoding process, detection speed could be truly unleashed. Thus, a speed-oriented concept emerges naturally: "Deep decoding, once is enough." The challenge, of course, lies in maintaining detection accuracy. Second, the lightweight gain (Lg) in SOD-RSI models remains ambiguous. The mismatch between model size and actual efficiency indicates the need for a strategy to effectively balance detection accuracy and computational efficiency.

Based on the above analysis, we propose a novel speed-oriented lightweight SOD-RSI network (SOLNet). "Deep decoding, once is enough" is the core idea of our method. First, considering the issue of accuracy loss during model compression, our detection framework does not rely on pretrained VGG or ResNet structures as the backbone. Instead, we employ a cascaded structure with multibranch feature fusion and train it from scratch. This is different from existing SOD-RSI works. Then, we propose a lightweight group attention (LGA) module to deconstruct–interact–recombine high-level feature maps. This design captures key information from deep channel features at a low computational cost. To improve the decoding accuracy, we further propose an enhanced dynamic encoding (EDE) module by dynamically weighting the convolution operator. Finally, a novel dynamically enhanced aggregation module (DEAM) is designed based on LGA and EDE, which reduces the complexity of the structure and achieves efficient decoding. Besides, to further analyze the performance of the SOD-RSI method, we propose a strategy to measure the model Lg. Our main contributions are summarized as follows.

1) We propose a DEAM to deconstruct–interact–recombine channel features and dynamically capture spatial information, which consists of the proposed LGA and EDE.
2) We propose SOLNet to achieve lightweight and efficient decoding, which uses the proposed DEAM to mine the intrinsic correlation of high-level feature information, eliminating the need to pay additional attention to other scales.
3) We propose a strategy to evaluate the Lg of models, further revealing the performance of different methods when considering both speed and accuracy.
4) Experimental results on the challenging SOD-RSI datasets show that our method achieves significant speed improvements and is the state-of-the-art (SOTA) lightweight SOD-RSI method.

II. Related Work

RSI exhibits complex spatial layouts and texture features, with larger variations in object scales, more frequent edge overlaps, and imbalanced class distributions compared to NSI. This challenge makes existing SOD-NSI methods [15], [16],

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[17], [18], [19], [20], [21] insufficient to meet the accuracy demands of the SOD-RSI task.

To this end, researchers have gradually shifted toward developing SOD-RSI models specifically designed for RSIs. In this context, CNNs have garnered significant attention in the SOD-RSI task due to their powerful and flexible feature extraction capabilities. CNNs can effectively capture local features in images and automatically learn different levels of feature representations through multiple convolutional layers, thereby adapting to the diversity and complexity of objects in RSIs. Specifically, the construction of CNN-based SOD-RSI models mainly involves three aspects: multiscale detection, capture of key information, and extraction of edge features. The mainstream method to solve the multiscale detection issue is to change the image resolution and design the network structure hierarchically. For example, LVNet [28], multiscale joint region and boundary model (MJRBM) [23], and adaptive edge-aware semantic interaction network (AESINet) [25] capture multiscale salient objects through the fusion of features at different levels and different resolutions. To address the issue of capturing key information, the mainstream method is to employ attention mechanisms that perceive global features. Specifically, DAFNet [29] captures long-distance semantic relationships by building a global context-aware attention module and further strengthens features through step-by-step low-high feature fusion. MSCNet [30] designs a parallel multiscale attention structure to remotely model the spatial and channel dimensions of features, thereby promoting accurate prediction results. ACCoNet [31] highlights and enhances salient regions by coordinating multilevel local and global features. As for the extraction of edge features, current methods mainly consider the constraints of the structure and loss function. Typically, an edge-aware multiscale feature integration network (EMFINet) [37] utilizes explicit and implicit cues of contour details to integrate multiscale features and performs edge-aware constraints through a hybrid loss function.

Aside from traditional CNN models, an increasing number of recent works [32], [33] have incorporated the ViT architecture into the SOD-RSI task. ViT demonstrates unique advantages in modeling long-range dependencies, enabling efficient capture of global context within images. The hybrid model combining ViTs and CNNs takes full advantage of the complementary advantages of long-distance feature relationship modeling and local detail extraction. However, the self-attention mechanism in ViTs requires processing the global relationships between all pixels, which causes computational complexity to increase quadratically with input size. This issue becomes especially pronounced in high-resolution RSIs. Moreover, the ViT requires large-scale datasets for training to fully exert its advantages, but the sparse data of SOD-RSIs further affects its effectiveness. As a result, most current hybrid models rely on fine-tuning pretrained weights to ensure satisfactory detection accuracy.

The above methods have achieved significant breakthroughs in accuracy, but they have not sufficiently considered the issues of computational cost and operational efficiency. Therefore, the lightweight SOD-RSI has become a new research hotspot [30], [38]. Most of the current lightweight methods come from the SOD-NSI task [19], [22], [34], [39], and few lightweight models are focusing on the SOD-RSI task. Even though recent studies such as CorrNet [35], SeaNet [1], and multiscale edge-embedded attention network (MEANet) [27] have reduced computational parameters by designing different lightweight structures, there is no gap in actual detection speed with ordinary SOD models such as DAFNet [29] and gated network (GateNet) [19].

Notably, recent works have compressed the computational parameters of models to around 1 M, showing improved speed in multibatch testing. However, the single-frame detection speed compared to models with parameters ranging from 10 to around 100 M has not significantly improved. In fact, the method of sparsifying and discretizing computational units to reduce computational parameters has not achieved the goal of GPU-friendly rapid inference. The inference speed of a model is influenced by two factors: parameter computation and data interaction. Existing methods usually focus on reducing model parameters to minimize computation time, while frequently overlooking the time required for data interaction. When the latter surpasses the normal computation time, an interesting phenomenon arises: fewer parameters may still result in higher inference latency. For example, the unoptimized replacement of standard convolutions with depthwise separable convolutions [40] to improve encoding efficiency can introduce challenges. Although depthwise separable convolutions reduce computational costs, they require more memory access operations than standard convolutions, increasing the demand on GPU memory bandwidth and making data transfer a performance bottleneck. In addition, some lightweight works reuse multilevel features for secondary encoding to maintain accuracy. Repeated encoding of multilevel features means that additional computing operations are required to complete the entire inference process, which will also increase the computational burden on the GPU and thus affect operating efficiency. Current works have mainly achieved lightweight at the model parameter level, while speed, as a key metric for evaluating model lightweight, still needs further optimization. Obviously, this has also become the motivation for us to delve into lightweight SOD-RSIs and then propose a new lightweight SOD-RSI network in this article, which truly achieves a leap in detection speed.

III. Approach

A. Framework Overview

The overall structure of the network is shown in Fig. 2. Our design concept is to encode efficiently and decode simply. Therefore, instead of following the design pattern of previous works that use VGG or ResNet as the backbone, we design the network encoder (S_0–S_4) as a RepVGG-style structure with multilevel receptive fields [41]. With the support of structural re-parameterization, this design achieves the training accuracy of a multibranch structure without adding inference costs. We use RepVGG-A0, the version with the fewest parameters in the RepVGG series, as the reference backbone and reshape the overall structure by scaling the channel ratio. Notably, our method does not rely on any pretrained parameters, giving us

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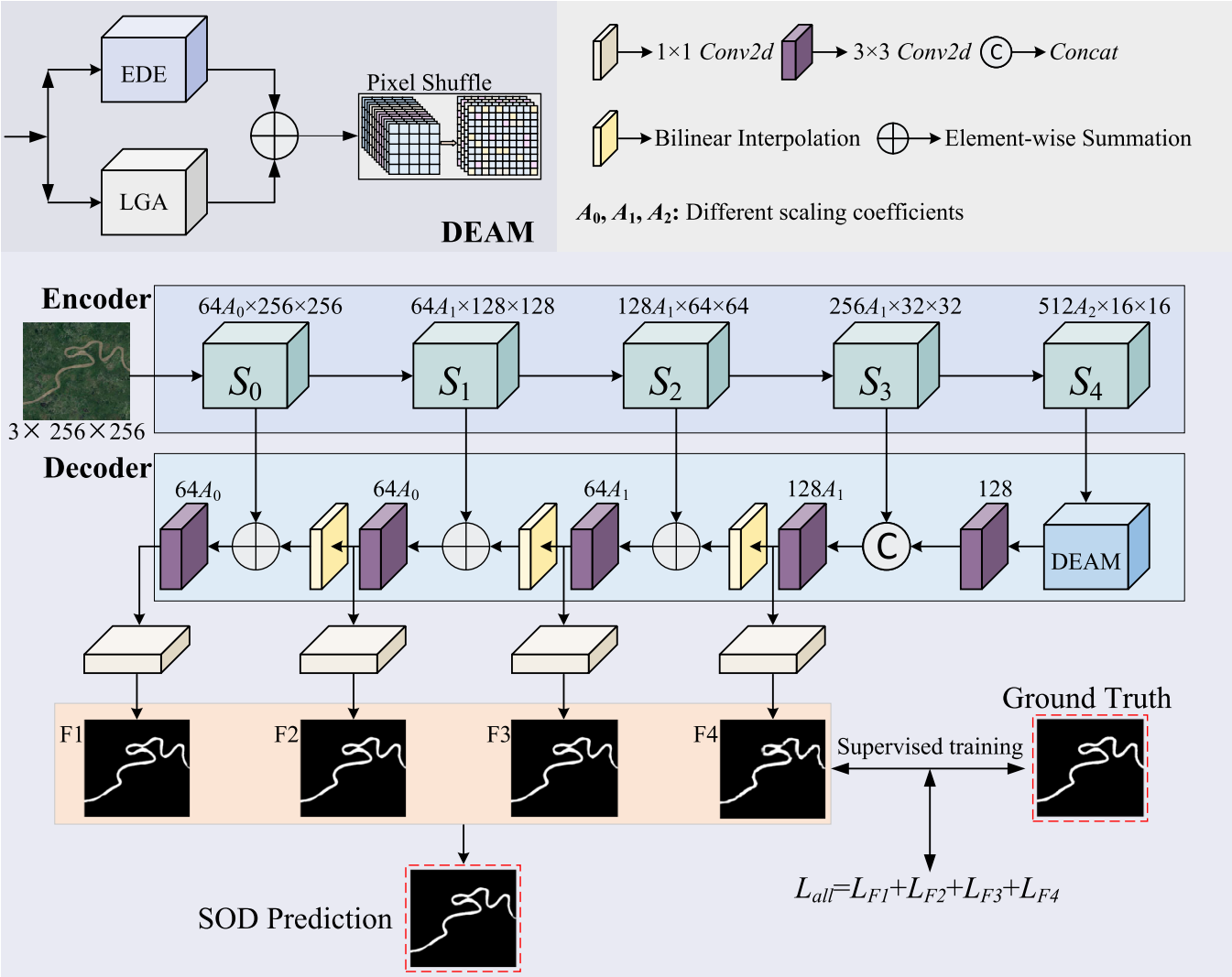


Fig. 2. Overall framework of the proposed method.

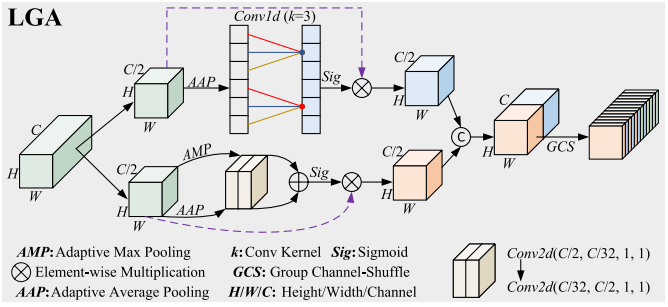


Fig. 3. Illustration of the proposed LGA.

the flexibility to design and optimize the network architecture according to the specific needs of the SOD-RSI task, free from the constraints of pretrained models. In contrast, recent SOD-RSI models based on transfer learning [1], [24], [25], [27], [31] do not have this advantage. Our analysis of existing models [1], [23], [25], [27], [31], [35], [36], [37] reveals that the high inference latency of low-parameter networks is mainly due to secondary encoding of multilevel features, which imposes a significant computational burden on the GPU. This means that these models may prioritize accuracy at the expense of efficiency. Thus, simplifying the decoding process while maintaining acceptable accuracy fluctuations is the key to improving inference speed.

In our method, we avoid adding additional operations on multiscale feature maps during decoding and instead focus on processing high-level features. Specifically, we design the EDE and LGA modules, respectively, and further propose the DEAM module, which only decodes high-level features, significantly reducing the decoding time. DEAM fully exploits the deep features of high-level feature maps by dynamically perceiving global and local information, thereby achieving high-precision decoding without processing low-level high-resolution feature maps. After fusing the two modules, we use Pixel Shuffle to upsample the feature maps. This not only facilitates subsequent feature association but also enhances information exchange across different channels and spatial features. Our design is based on three key considerations. First, dedicated modules that focus on enhancing specific types of features have obvious advantages over balancing multiple features in a single module. Second, separating spatial and channel information into two modules can provide a more flexible structure, thus facilitating the design of lightweight

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modules. Third, kernel operators are logically simpler and more straightforward than traditional spatial mechanisms.

Additionally, we use ground-truth labels during training for deep supervision at different prediction levels. The joint optimization of multiscale feature loss could prevent the training process from getting trapped in local optima, promote stable convergence, and improve prediction accuracy.

B. Lightweight and Efficient Decoding

Introducing attention mechanisms [42], [43], [44], [45], [46] that connect contextual information into the SOD-RSI network, especially the processing of multiscale feature maps, is advantageous for improving detection accuracy. However, current works [23], [25], [31] pay little attention to the issue of detection efficiency, and the design of network structure tends to be complicated.

To this end, we propose the LGA taking into account the calculation parameters of the unit module and the complexity of the overall network structure, as shown in Fig. 3. The motivation behind the LGA structure design includes two aspects: 1) fully exploiting the deep information of high-level feature maps and 2) modeling the feature dependencies in different dimensions with as few calculation parameters as possible. The similarity analysis of channel features in GhostNet [47], [48] shows that as the number of network coding layers and the number of feature channels increases, there are similarities between different channel features. Based on this research, we group channel features and then focus on features of different dimensions from different perspectives. This not only captures rich channel features but also effectively controls the computational cost. Specifically, LGA first divides the input features into two parts. One part of the feature undergoes AAP, followed by 1-D convolution to achieve local channel feature interaction without dimension reduction. The other part first compresses the feature dimensions through AMP and AAP and then adjusts the dimensions through a set of 2-D convolutions. The two parts of features are fused in the channel dimension after being activated by the Sig function. Since they focus on different information, we further apply GCS on the fused features to fully realize feature interaction in the channel dimension. The complete calculation process is as follows:

$$\left(S_4', S_4''\right) = \text{Split}(S_4) \tag{1}$$

$$f(\text{Conv}, \text{op}, x, k) = \text{Conv}(\text{op}(x), k) \tag{2}$$

$$S_4^{(1)} = S_4' \otimes \text{Sig}\left(f\left(\text{Conv1d}, \text{AAP}, S_4', 3\right)\right) \tag{3}$$

$$S_4^{(2)} = S_4'' \otimes \text{Sig}\left(f\left(\text{Conv2d}'', \text{AAP}, S_4'', 1\right) + f\left(\text{Conv2d}'', \text{AMP}, S_4'', 1\right)\right) \tag{4}$$

$$S^{(3)} = \text{GCS}\left(\text{Concat}\left(S_4^{(1)}, S_4^{(2)}\right), 4\right) \tag{5}$$

where Split is a channel separation operation and S_4' and S_4'', respectively, represent the two parts of features after separation. Conv1d represents a 1-D convolution operation and Conv2d$''$ represents a set of convolutions used to adjust dimensions in Fig. 3. GCS represents the channel shuffle operation and 4 represents the number of groupings for channel shuffle. Especially, to simplify memory and clearly describe the calculation process, we record the second-order encoding method of the feature map as $f(\text{Conv}, \text{op}, x, k)$. Conv represents the convolution operation, op represents the first-order encoding operation (op $= 0$ means no first-order encoding), x represents the input, and k represents the convolution kernel.

Existing methods for capturing spatial information mainly rely on applying spatial weights to feature maps through additional modules. In our view, directly integrating the spatial information modeling process into the convolution kernel computation could eliminate the need for weighted calculation modules. Based on this concept, we design the EDE module to adaptively capture spatial information, as shown in Fig. 4. Different from spatial self-attention mechanisms [49], we pay more attention to how to use the coding characteristics of CNNs to dynamically capture information of interest. Currently, some works studying the encoding form of convolution operators [50], [51], [52], [53] focus on hierarchical weighting of kernels to improve accuracy, while paying less attention to computational cost. Therefore, we re-examine the computation process of kernel operators to seek a simpler method.

First, based on the exploration of LGA, we perform channel interaction on the kernel operator without reducing the channel dimension, as shown as follows:

$$f^1 = f(\text{Conv1d}, \text{AAP}, S_4, k). \tag{6}$$

Then, the kernel operator is deeply decomposed from the input channel features, output channel features, and the number of kernels. The spatial information modeling capability of a 2-D convolution operator is closely related to the size of the kernel. A larger kernel increases the receptive field but also raises the computational cost, while a smaller kernel reduces the receptive field, which weakens its spatial modeling capabilities. Using large kernels could capture broad spatial dependencies, but as the number of feature channels grows, the computational complexity of these methods increases significantly, imposing higher demands on hardware capabilities. Therefore, instead of following this way, we further refined the kernel encoding process. Specifically, as shown in parts I–III in Fig. 4, we shift the postprocessing operations originally applied to the feature map to the kernel encoding stage. By adding additional dimensions to focus on the number of kernels, we enhance the extraction of spatial features from the image. In fact, different dimensions require varying levels of attention, resulting in the extraction of distinct feature information. Using fixed-weight coefficients to balance the three parts would necessitate numerous repeated experiments to fine-tune the parameters, without ensuring that the chosen coefficients are optimal. A more effective method is to allow the coefficients to be adjusted adaptively, where their gradients can be updated during the training process, as shown below:

$$\begin{cases} w^1 = w_1 * \text{Sig}\left(f^1\right) \\ w^2 = w_2 * \text{Sig}\left(f\left(\text{Conv2d}^{C_{\text{out}}}, 0, f^1, 1\right)\right. \\ w^3 = w_3 * \text{Sfm}\left(f\left(\text{Conv2d}^n, 0, f^1, 1\right)\right. \end{cases} \tag{7}$$

Different kernel attention matrices are generated through dynamic weighting of input, output, and number of kernels. w_1, w_2, and w_3, respectively, represent the learnable weights of the kernel operator in different dimensions. Conv2d$^{C_{\text{out}}}$ and

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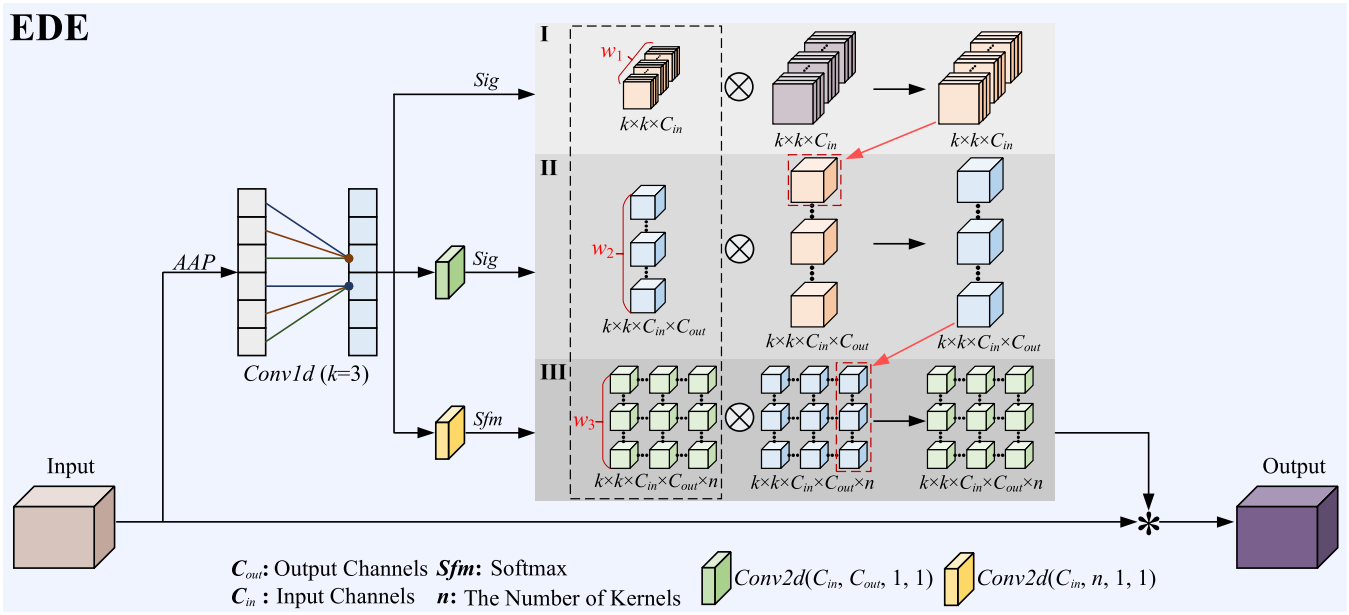


Fig. 4. Structural diagram of the proposed EDE. I represents multiplication along the input channel dimension, II represents multiplication along the output channel dimension, and III represents multiplication along the spatial kernel dimension of the convolution filters. w_1, w_2, and w_3 denote the learnable weight coefficients corresponding to each respective operation. The weighted contributions from I to III are related to the input. $*$ indicates convolution operation. After weighting, the convolution kernel encodes the input to produce the output.

Conv2d^n are used to adjust dimensions in different stages. C_{out} represents the output channels and n represents the number of kernels.

Finally, the three attention matrices are multiplied in sequence to generate the dynamic fusion kernel operator k^e. Compared with traditional static convolution, k^e integrates feature dependencies in multiple dimensions. In particular, the added dimension of focusing on the number of kernels enables it to dynamically generate the most suitable kernel for each sample in multiple dimensions, thereby improving the expressiveness and adaptability of the model. The enhanced feature map is denoted as f^e as follows:

$$\begin{cases} k^e = \sum_{d=1}^{n} \prod_{j=1}^{3} w^j k_d \\ f^e(\text{Conv2d}, 0, S_4, k) = f\left(\text{Conv2d}, 0, S_4, k^e\right) \end{cases}. \tag{8}$$

Notably, we do not specifically emphasize spatial information within the kernel for two reasons. First, spatial position encoding is an intrinsic feature of 2-D convolutions, and the dynamic enhancement of kernels across multiple dimensions already improves spatial information perception. Second, our primary focus is on minimizing computational complexity. Cascade weighting of feature matrices can lead to exponential growth in computational cost, so we reduce complexity by altering the order of weighting calculations.

C. Supervision Strategy

Our supervision strategy mainly includes three parts: training data enhancement, loss function, and Lg evaluation.

1) Data Enhancement: Limited image data for SOD-RSIs makes most works use pretrained networks as the backbone for fine-tuning. However, this limits the design of network structures specific to the SOD-RSI task. To train a lightweight network for SOD-RSI from scratch, we first integrate the existing SOD-RSI data and then expand these data by rotating, symmetry, cropping, and adding noise. With this strategy, the number of image data has been expanded from the original thousands to tens of thousands, breaking free from the limitations of pretrained network structures and laying the data foundation for custom lightweight structures.

2) Loss Function: In the supervised training illustrated in Fig. 2, the overall training loss L_{all} is composed of prediction losses at different scales. We set up three supervisions to collaboratively optimize the losses. The definitions are as follows:

$$L_{\text{all}} = \sum_{i=1}^{4} \left(L_{Fi}^{\text{BCE}} + L_{Fi}^{\text{IOU}} + L_{Fi}^{M}\right) \tag{9}$$

where L^{BCE}, L^{IOU}, and L^M represent binary cross-entropy loss, intersection over union (IOU) loss, and the harmonic loss of precision and recall, respectively. $F1$ represents the $F1$ feature map in Fig. 2. The specific definitions are as follows:

$$L^{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \begin{bmatrix} S_t(i) \log S_p(i) \\ +(1 - S_t(i)) \log\left(1 - S_p(i)\right) \end{bmatrix} \tag{10}$$

$$L^{\text{IOU}} = 1 - \frac{\sum_{i=1}^{N} \left(S_t(i) S_p(i)\right)}{\sum_{i=1}^{N} \left(S_t(i) + S_p(i) - S_t(i) S_p(i)\right)} \tag{11}$$

$$\begin{cases} L^M = 1 - \dfrac{\left(1 + \beta^2\right) PR}{\beta^2 P + R} \\ P = \dfrac{\sum_{i=1}^{N} \left(S_t(i) S_p(i)\right)}{\sum_{i=1}^{N} S_p(i)}, R = \dfrac{\sum_{i=1}^{N} \left(S_t(i) S_p(i)\right)}{\sum_{i=1}^{N} S_t(i)} \end{cases} \tag{12}$$

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where S_t represents the label, S_p represents the predicted saliency map, and N represents the total number of pixels. P represents precision and R represents recall. β^2 is the harmonic factor, which we set as 0.3. It can be observed from the function definitions of the IOU loss and the harmonic loss that the latter further strengthens the constraint on precision based on the former. Note that we only add L_{F1}^{M} as an additional value to the losses at different scales. This mitigates the negative impact of decreased constraint caused by weighted summation.

3) Lg Evaluation: The lightweighting of SOD-RSI models is an emerging topic at present. Existing works do not propose a reasonable evaluation strategy to analyze speed and accuracy simultaneously. Based on this, to comprehensively analyze the performance of lightweight models, we propose an evaluation strategy that takes into account both, called "Lg." For a convincing evaluation strategy, the contribution of different factors to the results should be similar. The more similar the contributions of different factors are, the more accurate the evaluated results will be. To scientifically and rigorously measure the Lg of the model, we analyze the test results of 23 relevant works [1], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [27], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38] from the past three years in terms of speed and accuracy. The analysis results show that the difference in accuracy between different methods is between 0.1 and 1, while the difference in frames per second (FPS) is between 10 and 100. Therefore, we define the Lg as the product of the FPS scaled down by 100 times and the accuracy value, as follows:

$$\mathrm{Lg} = \frac{v}{100} \times \frac{\sum_{m=1}^{M} \mathrm{acc}_i}{M} \tag{13}$$

where v represents FPS, acc_i represents the accuracy of the ith category metric, and M represents the total number of categories of metrics. Using the proposed Lg metric, the tradeoff results between the speed and accuracy of different methods could be clearly analyzed.

IV. Experiments

A. Experimental Setting

1) Dataset: Experiments are conducted on two public challenging SOD-RSI datasets: ORSSD [28] and EORSSD [29].

The ORSSD dataset comprises aerial imagery collected from Google Earth and various public remote sensing sources. These images are manually filtered to include surface views of urban, rural, desert, and forest areas. The dataset features a diverse range of natural and man-made objects, such as shadows, trees, buildings, airplanes, ships, cars, rivers, ponds, bridges, stadiums, beaches, deserts, and dense forests. It consists of 800 images with varying spatial resolutions, with 600 images designated for training and 200 for testing.

The EORSSD dataset extends the ORSSD dataset by incorporating additional images extracted from Google Earth, which include both high-resolution satellite imagery and low-resolution aerial imagery. This dataset covers a wide range of scenes and includes nine distinct remote sensing object types: ships, airplanes, cars, buildings, water, islands, roads, none, and others. It comprises 2000 images with varying spatial resolutions, with 1400 images allocated for training and 600 for testing.

To train SOLNet from scratch, we merge the training sets of the two datasets and expand the dataset to 28 000 images by rotating, flipping, color enhancement, adding Gaussian noise, cropping, and so on.

2) Implementation Details: The network backbone adopts the RepVGG-A0 structure [41], and the channel coefficients (A_0, A_1, A_2) are adjusted to $(0.5, 0.75, 1)$. The batch size is 32 and the input image is resized to 256^2. Currently, in deep learning-based SOD methods, the input image resolution typically ranges between 128^2 and 448^2. By observing the performance of existing works at different resolutions, we set the input resolution to 256^2 to balance computational efficiency and detection accuracy. We train SOLNet for 120 epochs using the ADAM optimizer [54], with an initial learning rate of $1e^{-4}$, a decay factor of 0.1, and a decay step of 80. All experiments are conducted on an RTX 3080 Ti GPU, and the complete training process takes approximately 10 h.

B. Metrics

We adopt ten evaluation metrics to comprehensively evaluate the performance of the model. There are six accuracy evaluation metrics, including F-measure ($\overline{F_\beta}$, F_β^{adp}) [55], E-measure ($\overline{E_\xi}$, E_ξ^{adp}) [56], mean absolute error ($\mathcal{M}$) [57], and S-measure (S_α) [58]. $\overline{F_\beta}$ and $\overline{E_\xi}$, respectively, represent the mean scores under the corresponding metrics. F_β^{adp} and E_ξ^{adp} represent the adaptive threshold scores under the corresponding metrics. The details are as follows:

$$F_\beta = \frac{\left(1+\beta^2\right) \times P \times R}{\beta^2 \times P + R} \tag{14}$$

$$E_\xi = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} \theta(\xi) \tag{15}$$

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} |S_p(i) - S_t(i)| \tag{16}$$

$$S_\alpha = 0.5 \times \left(S_o\left(S_p, S_t\right) + S_r\left(S_p, S_t\right)\right) \tag{17}$$

where H and W represent the height and width of the saliency map, respectively, ξ is the alignment matrix, $\theta(\xi)$ denotes the enhanced alignment matrix, and S_o and S_r denote object-aware and region-aware structural similarity.

There are four speed metrics, which are network computational parameters (Params), floating-point operations (FLOPs), inference speed reported in the FPS unit, and Lg. The inference speed is reported under the conditions of batch size 1 and only includes model forward inference.

C. Qualitative and Quantitative Performance Comparison

We conduct comparative experiments between the proposed SOLNet and 23 related methods [1], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [27], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38] in the fields of SOD-NSIs and

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TABLE I

QUANTITATIVE COMPARISON OF DIFFERENT METHODS UNDER MULTIPLE METRICS ON THE EORSSD AND ORSSD DATASETS. ↑ MEANS THE HIGHER THE BETTER, AND ↓ MEANS THE SMALLER THE BETTER. OSN REPRESENTS THE ORDINARY SOD-NSI METHOD, OSR REPRESENTS THE ORDINARY SOD-RSI METHOD, LSN REPRESENTS THE LIGHTWEIGHT SOD-NSI METHOD, AND LSR REPRESENTS THE LIGHTWEIGHT SOD-RSI METHOD

Type	Methods	Years	EORSSD						ORSSD					
			$S_\alpha \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^{adp} \uparrow$	$\overline{E_\xi} \uparrow$	$F_\beta^{adp} \uparrow$	$\overline{F_\beta} \uparrow$	$S_\alpha \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^{adp} \uparrow$	$\overline{E_\xi} \uparrow$	$F_\beta^{adp} \uparrow$	$\overline{F_\beta} \uparrow$
OSN	GCPA [17]	2020	0.8869	0.0102	0.8647	0.9167	0.6723	0.7905	0.9026	0.0168	0.9205	0.9341	0.7861	0.8433
	MINet [16]	2020	0.9040	**0.0093**	0.9243	0.9346	0.7705	0.8174	0.9040	0.0144	0.9423	0.9454	0.8251	0.8574
	ITSD [15]	2020	0.8050	0.0106	0.9103	0.9407	0.7421	0.8221	0.9050	0.0165	0.9335	0.9482	0.8068	0.8502
	GateNet [19]	2020	0.9114	0.0095	0.8909	0.9385	0.7109	0.8228	0.9186	**0.0137**	0.9428	0.9538	0.8229	0.8679
	SUCA [20]	2021	0.8988	0.0097	0.9082	0.9277	0.7260	0.7949	0.8989	0.0145	0.9194	0.9400	0.7748	0.8237
	PA-KRN [21]	2021	**0.9192**	0.0104	**0.9416**	**0.9536**	**0.7993**	**0.8358**	**0.9239**	0.0139	**0.9579**	**0.9620**	**0.8548**	**0.8727**
OSR	DAFNet [29]	2021	0.9166	0.0060	0.8446	0.9291	0.6427	0.7845	0.9191	0.0113	0.9360	0.9539	0.7876	0.8511
	MJRBM [23]	2022	0.9197	0.0099	0.8897	0.9350	0.7066	0.8239	0.9204	0.0163	0.9328	0.9415	0.8022	0.8566
	EMFINet [37]	2022	0.9290	0.0084	0.9501	0.9604	0.7984	0.8486	0.9366	0.0109	0.9663	0.9671	0.8617	0.8856
	MCCNet [36]	2022	0.9327	0.0066	0.9538	0.9685	**0.8137**	0.8604	0.9437	0.0087	0.9735	0.9758	**0.8957**	**0.9054**
	AESINet [25]	2023	0.9358	0.0079	0.9462	0.9636	0.7923	0.8524	0.9460	0.0086	0.9707	0.9747	0.8666	0.8986
	AccoNet [31]	2023	0.9290	0.0074	0.9450	0.9653	0.7969	0.8552	0.9437	0.0088	0.9721	0.9754	0.8806	0.8971
	ERPNet [24]	2023	0.9210	0.0089	0.9228	0.9401	0.7554	0.8304	0.9254	0.0135	0.9520	0.9566	0.8356	0.8745
	HFCNet [33]	2024	**0.9449**	**0.0049**	**0.9552**	**0.9720**	0.8051	**0.8646**	**0.9543**	**0.0070**	**0.9748**	**0.9798**	0.8710	0.9049
	TLCKDNet [32]	2024	0.9350	0.0056	0.9514	0.9661	0.7969	0.8535	0.9421	0.0082	0.9696	0.9710	0.8719	0.8947
LSN	CSNet [18]	2020	0.8364	0.0169	**0.8339**	**0.8929**	**0.6319**	**0.7656**	**0.8910**	**0.0186**	**0.9068**	**0.9171**	**0.7615**	**0.8285**
	SAMNet [22]	2021	0.8622	0.0132	0.8284	0.8700	0.6114	0.7214	0.8761	0.0217	0.8656	0.8818	0.6843	0.7531
	HVPNet [34]	2021	**0.8734**	**0.0110**	0.8270	0.8721	0.6202	0.7377	0.8610	0.0225	0.8471	0.8717	0.6726	0.7396
LSR	CorrNet [35]	2022	0.9289	0.0083	0.9593	0.9646	0.8311	0.8620	0.9380	0.0098	0.9721	0.9746	0.8875	0.9002
	MSCNet [30]	2022	0.9078	0.0091	0.9257	0.9533	0.7444	0.8138	0.9233	0.0130	0.9554	0.9648	0.8323	0.8670
	FSMINet [38]	2022	0.9255	0.0079	0.9490	0.9567	0.8015	0.8436	0.9361	0.0101	0.9693	0.9672	0.8710	0.8878
	SeaNet [1]	2023	0.9208	0.0073	0.9602	0.9651	0.8304	0.8519	0.9260	0.0105	0.9670	0.9722	0.8625	0.8772
	MEANet [27]	2024	0.9282	0.0070	0.9663	0.9658	0.8557	0.8678	0.9340	0.0098	0.9715	0.9730	0.8860	0.8934
	SOLNet (Ours)	-	0.9171	0.0078	0.9596	0.9577	0.8392	0.8513	0.9284	0.0111	0.9697	0.9689	0.8925	0.8946

Bold data represents the best results in the corresponding category. The top three LSR results are marked red, blue and green respectively.

TABLE II

COMPARISON OF MODEL SIZE AND RUNNING TIME OF DIFFERENT METHODS. LG(EOR) AND LG(OR) REPRESENT THE LG UNDER THE EORSSD AND ORSSD DATASETS, RESPECTIVELY

Type	Methods	Years	Params (M)	FLOPs (G)	Speed (FPS)	Lg (EOR)	Lg (OR)
OSN	GCPA [17]	2020	67.1	**54.4**	80	0.54	0.56
	MINet [16]	2020	47.6	146.4	62	0.43	0.44
	ITSD [15]	2020	**16.7**	57.4	75	0.65	0.68
	GateNet [19]	2020	100.0	108.4	**99**	**0.87**	**0.91**
	SUCA [20]	2021	115.6	56.8	69	0.60	0.61
	PA-KRN [21]	2021	141.1	617.7	22	0.20	0.20
OSR	DAFNet [29]	2021	**29.4**	68.5	**86**	**0.73**	**0.78**
	MJRBM [23]	2022	43.5	95.7	42	0.37	0.38
	EMFINet [37]	2022	95.1	176.9	36	0.33	0.34
	MCCNet [36]	2022	67.7	117.2	65	0.60	0.62
	AESINet [25]	2023	41.1	**53.6**	14	0.13	0.13
	AccoNet [31]	2023	102.6	184.5	58	0.53	0.55
	ERPNet [24]	2023	56.5	87.3	77	0.69	0.71
	HFCNet [33]	2024	148.99	160.71	16	0.15	0.15
	TLCKDNet [32]	2024	52.09	31.72	35	0.32	0.33
LSN	CSNet [18]	2020	**0.1**	0.7	56	0.46	0.49
	SAMNet [22]	2021	1.3	**0.6**	41	0.33	0.34
	HVPNet [34]	2021	1.2	1.2	**62**	**0.51**	**0.51**
LSR	CorrNet [35]	2022	4.1	21.3	82	0.76	0.77
	MSCNet [30]	2022	3.3	5.9	74	0.66	0.68
	FSMINet [38]	2022	3.6	11.8	37	0.34	0.35
	SeaNet [1]	2023	**2.8**	**1.8**	72	0.66	0.67
	MEANet [27]	2024	3.3	11.3	31	0.29	0.29
	SOLNet (Ours)	-	6.5	8.1	**161**	**1.48**	**1.51**

Bold data represents the best results in the corresponding category.

SOD-RSIs. The superiority of SOLNet is intuitively revealed through comprehensive qualitative and quantitative analysis.

1) Quantitative Accuracy Evaluation: To illustrate the effectiveness of SOLNet, we evaluate the accuracy of 23 methods on the ORSSD and EORSSD datasets. The test results are shown in Table I. SOLNet shows quite competitive results on six metrics compared with the SOTA models of LSR and OSR in Table I. Especially, on the EORSSD dataset, SOLNet shows an improvement of 0.46% compared to the best E_ξ^{adp} achieved in OSR. On the ORSSD dataset, SOLNet exhibits a 0.56% improvement compared to the best F_ξ^{adp} achieved in LSR. These results demonstrate that the perception module we designed to capture local and global information is effective, which could balance pixel-level errors and optimize features. Objectively, due to the simplification of the network structure, SOLNet inevitably suffers from a loss of accuracy. On the ORSSD and EORSSD datasets, SOLNet scores slightly below the best results on some metrics. However, it remains the top three accuracy methods in LSR. While MEANet shows an accuracy advantage in Table I, it ranks last in the speed test shown in Table II. In contrast, SOLNet achieves significant speed improvements at the expense of negligible accuracy loss, achieving a favorable accuracy-efficiency tradeoff.

2) Quantitative Speed Comparison: We test the computational complexity and actual running time of 23 methods on the EORSSD dataset. To ensure a fair comparison, speed verification uses the methods of officially published codes and utilizes the same computational device for testing. Specifically, we also categorize these methods into four classes: OSN, OSR, LSN, and LSR. From the test results in Table II, the existing SOTA models of LSN and LSR have lower Params and FLOPs than the SOTA models of OSN and OSR, effectively compressing the size of the model. However, as discussed in the introduction regarding existing lightweight SOD models, it is evident that LSN and LSR do not achieve significant improvements in the golden benchmark of performance-speed, despite being lightweight models that are expected to provide performance enhancements. On the contrary, our proposed SOLNet, compared with OSN and OSR, has achieved a real leap in speed while significantly reducing Params and

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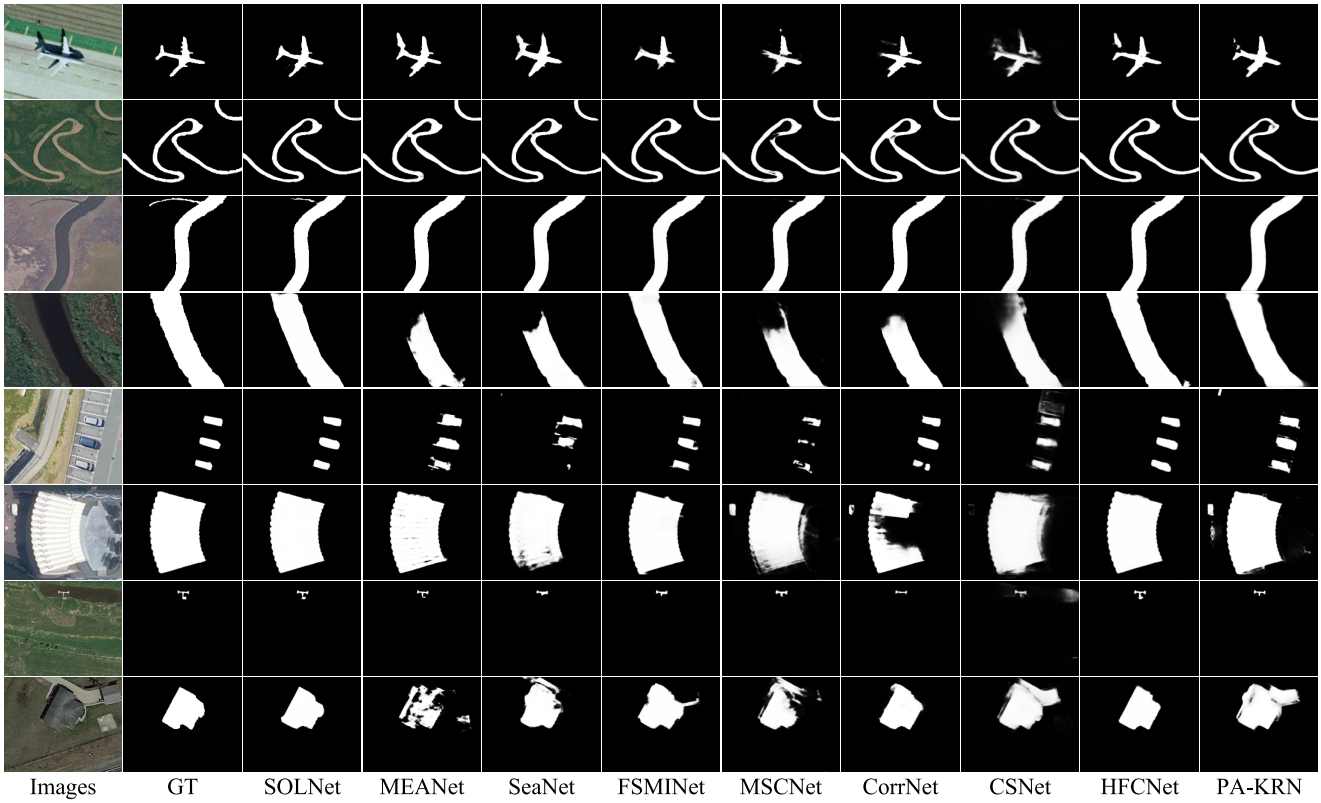


Fig. 5. Qualitative visual comparison with SOTA methods.

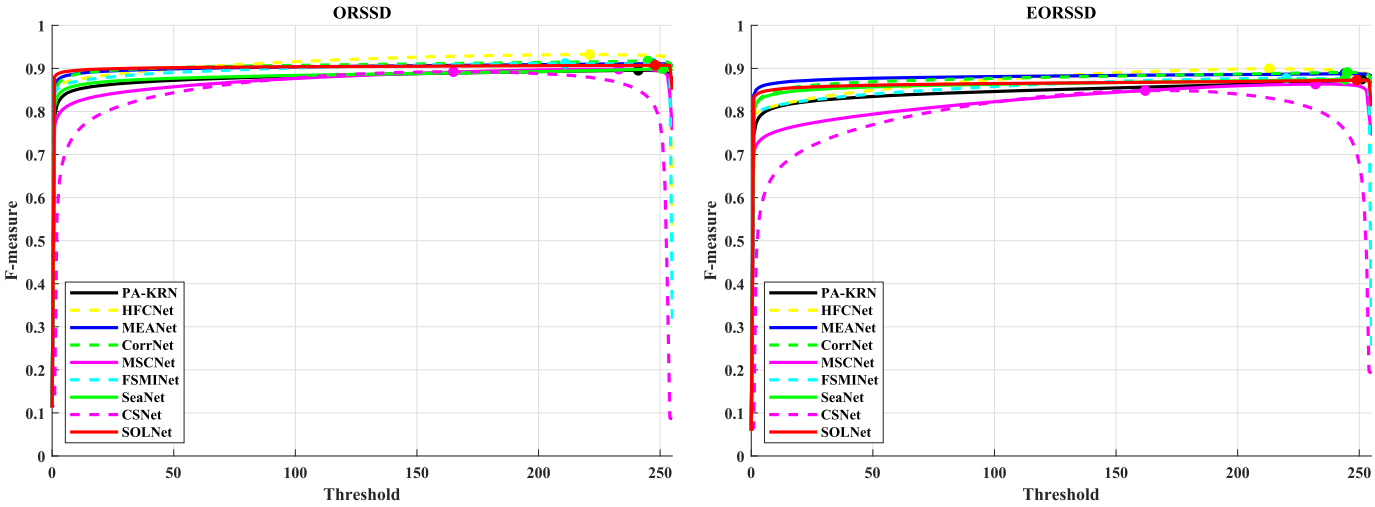


Fig. 6. F-Measure scores of different methods at different thresholds on the EORSSD and ORSSD datasets.

FLOPs. In terms of inference speed, SOLNet shows a 187.50% improvement over the most parameter-efficient model, CSNet. Furthermore, SOLNet achieves a 96.34% speed improvement compared to CorrNet, the fastest performing method in LSR. Not only that, compared with MEANet, the inference speed of SOLNet has been improved by an astonishing 419.35%, showing strong competitiveness. The detection speed of SOLNet 161 frames/s is far ahead of all methods, which demonstrates that our lightweight design for the network is feasible.

To comprehensively analyze the speed and accuracy of the model, we further test the Lg of different methods. As shown in Table II, the proposed method achieves the best scores in terms of Lg on both datasets. Strikingly, SOLNet attains approximately double the Lg compared to the second-ranked CorrNet in LSR. In previous works, GateNet obtained advanced results in both inference speed and Lg, but there is still a significant gap compared to the proposed method. On the EORSSD and ORSSD datasets, SOLNet outperforms GateNet with Lg increased by 70.11% and 65.93%, respectively.

Notably, the Lg of GateNet and DAFNet are higher than specialized lightweight methods, indicating that current

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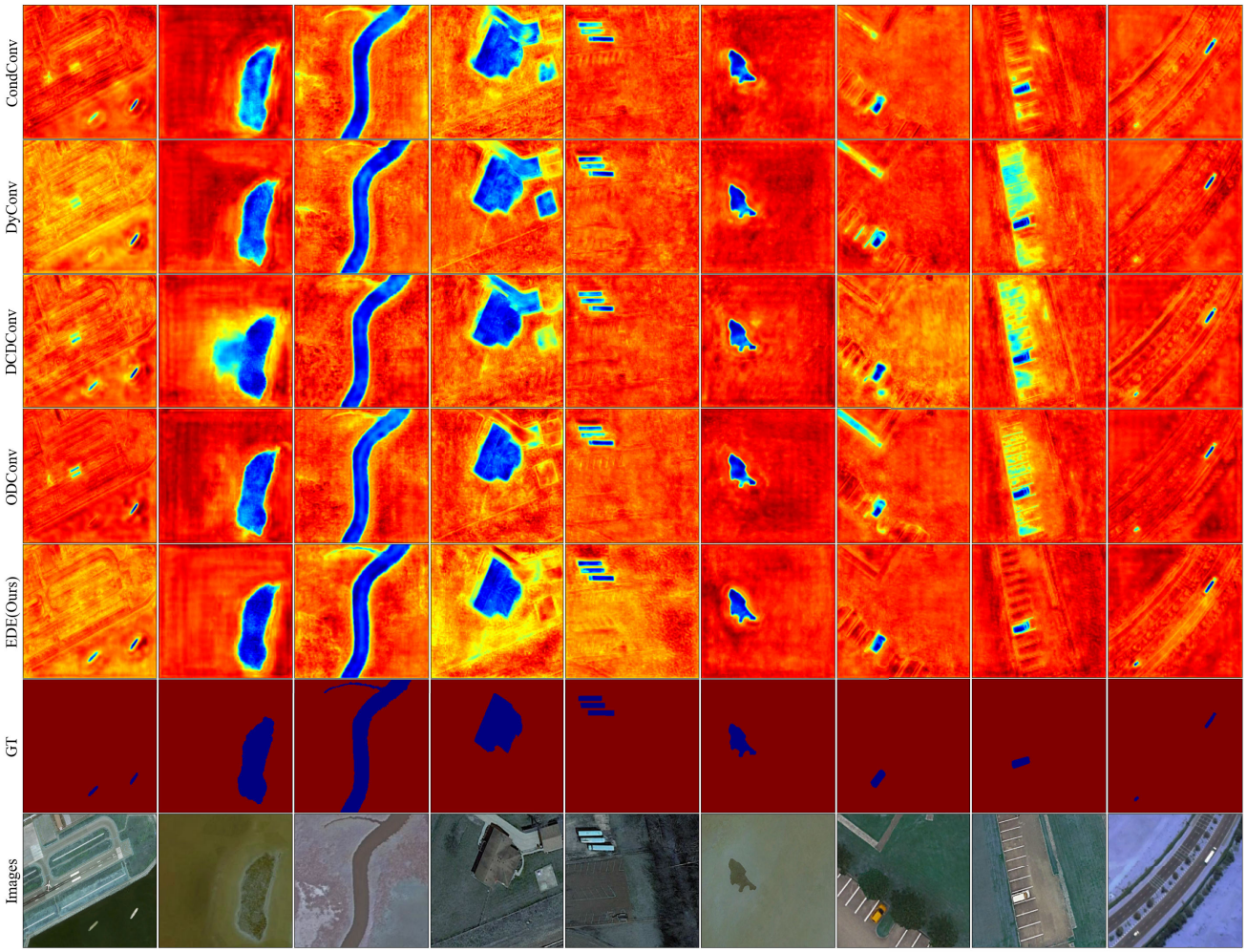


Fig. 7. Heatmaps of spatial information extracted by different modules.

TABLE III

ABLATION STUDY OF THE PROPOSED LGA AND EDE. THE FIRST PART AND THE SECOND PART ARE THE PERFORMANCE TESTS OF THE EDE AND LGA, RESPECTIVELY

Type	EORSSD						ORSSD					
	$S_\alpha \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^{adp} \uparrow$	$\overline{E_\xi} \uparrow$	$F_\beta^{adp} \uparrow$	$\overline{F_\beta} \uparrow$	$S_\alpha \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^{adp} \uparrow$	$\overline{E_\xi} \uparrow$	$F_\beta^{adp} \uparrow$	$\overline{F_\beta} \uparrow$
CondConv [50]	0.9087	0.0083	0.9535	0.9548	0.8254	0.8389	0.9242	0.0115	0.9652	0.9653	0.8808	0.8859
DyConv [51]	0.9113	0.0083	0.9527	0.9508	0.8311	0.8402	0.9266	**0.0110**	0.9672	0.9650	0.8881	0.8884
DCDConv [52]	0.9113	0.0089	0.9542	0.9532	0.8305	0.8407	0.9254	0.0116	0.9678	0.9674	0.8841	0.8876
ODConv [53]	0.9155	0.0089	0.9552	0.9556	0.8375	0.8478	**0.9290**	0.0113	0.9686	0.9674	0.8908	0.8914
EDE (Ours)	**0.9171**	**0.0078**	**0.9596**	**0.9577**	**0.8392**	**0.8513**	0.9284	0.0111	**0.9697**	**0.9689**	**0.8925**	**0.8946**
SE [42]	0.9002	0.0098	0.9367	0.9423	0.7974	0.8183	0.9168	0.0143	0.9571	0.9577	0.8646	0.8740
CBAM [43]	0.9100	0.0097	0.9475	0.9463	0.8297	0.8394	0.9250	0.0142	0.9654	0.9634	**0.8941**	0.8906
ECA [44]	0.9106	0.0092	0.9551	0.9507	0.8329	0.8415	0.9273	0.0113	0.9659	0.9637	0.8898	0.8914
SA [45]	0.9082	0.0089	0.9476	0.9450	0.8182	0.8306	0.9263	0.0112	0.9685	0.9675	0.8832	0.8869
EMA [46]	0.9123	0.0089	0.9564	0.9523	0.8372	0.8465	0.9234	0.0114	0.9651	0.9634	0.8909	0.8899
LGA (Ours)	**0.9171**	**0.0078**	**0.9596**	**0.9577**	**0.8392**	**0.8513**	**0.9284**	**0.0111**	**0.9697**	**0.9689**	0.8925	**0.8946**

Bold data represents the best results in the corresponding type.

lightweight SOD-RSI works suffer from under-gain issues. This also reflects the importance of our research.

3) Qualitative Accuracy Evaluation: Based on the evaluation results in Table I, we select all LSR methods [1], [27], [30], [35], [38] and PA-KRN [21], HFCNet [33], and CSNet [18], which performed best in their respective types, for further qualitative comparison. We provide visual results of the above methods to intuitively evaluate the detection performance of SOLNet. Taking into account factors such as target scale, target motion state, appearance shape, and environmental interference, the scenes in Fig. 5 include airplanes, rivers, adjacent cars driving on roads, interference from objects of the same color, and suburban scenes. The visual results in Fig. 5 demonstrate that compared to the current SOTA methods, SOLNet provides predictions for objects of different scales and irregular edge shapes that are closer to the GT, showing significant advantages in modeling object appearances. Additionally, in Fig. 6, we analyze the F-measure

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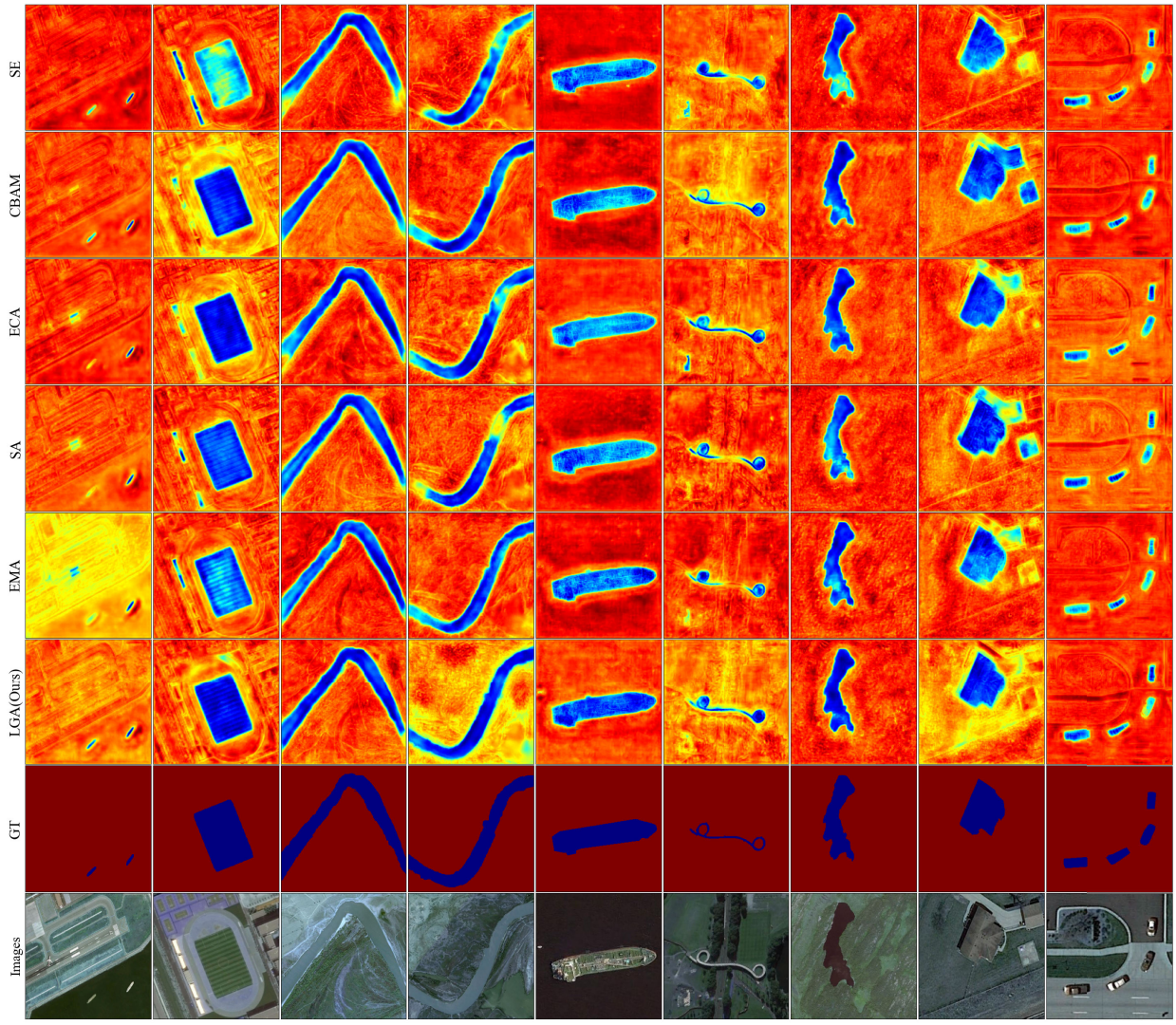


Fig. 8. Visual comparison of different attention modules.

scores of different methods under different thresholds. SOLNet demonstrates reliable accuracy and stability in F-measure curves on both datasets.

D. Ablation Study

Our ablation study on SOLNet is divided into two parts, including a horizontal comparison of each module and a reasonable assessment of the overall network structure.

1) Horizontal Comparison of Module Performance: As shown in Table III, we conduct a horizontal comparative test on the proposed LGA and EDE. For the EDE module, EDE performs better on the two test datasets than other methods. Although the best S_α and $\mathcal{M}$ are not achieved on the ORSSD dataset, the difference of less than 0.1% also demonstrates the tradeoff between accuracy and efficiency of EDE.

To further compare the ability to capture spatial information, we show heatmaps of the prediction results of different modules in challenging scenarios. Fig. 7 mainly includes three types of scenes: the target occupies a small proportion in the background, the contrast between the target and the environment is low, and the target appearance shape is slender. As shown in Fig. 7, SOLNet perceives the shape of objects more accurately than other modules. Especially, it provides prediction results that are more consistent with human eye attention in terms of capturing the small tubular branch structure of rivers, the degree of attention to weak and small targets, and the segmentation of objects and backgrounds under low contrast.

For the LGA module, LGA scores significantly higher on both datasets compared to the lightweight SA and SOTA methods EMA. The overall performance of LGA is perceived as satisfactory, mainly due to the profound interaction of feature information on channel dimensions.

Furthermore, to intuitively explain the superiority of LGA in object detail perception, we also provide heatmaps of different modules in multiple scenarios. The visualization results in Fig. 8 indicate that compared to other attention modules the designed LGA has stronger object feature representation capability and a more focused region of interest. For example, in scenarios such as the slender tubular structures of rivers, the irregular edge shapes of lakes, and the perception of nearby small targets, the prediction results of LGA are more

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TABLE IV
ABLATION STUDY OF THE PROPOSED NETWORK STRUCTURE AND LOSS FUNCTION

Tag	EORSSD						ORSSD					
	$S_\alpha \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^{adp} \uparrow$	$\overline{E_\xi} \uparrow$	$F_\beta^{adp} \uparrow$	$\overline{F_\beta} \uparrow$	$S_\alpha \uparrow$	$\mathcal{M} \downarrow$	$E_\xi^{adp} \uparrow$	$\overline{E_\xi} \uparrow$	$F_\beta^{adp} \uparrow$	$\overline{F_\beta} \uparrow$
w/o LGA+EDE	0.8981	0.0096	0.9445	0.9377	0.8148	0.8235	0.9132	0.0138	0.9605	0.9601	0.8737	0.8763
w/o LGA	0.9092	0.0087	0.9541	0.9468	0.8373	0.8428	0.9217	0.0119	0.9661	0.9625	0.8894	0.8883
w/o EDE	0.9121	0.0090	0.9496	0.9525	0.8289	0.8434	0.9265	0.0111	0.9680	0.9675	0.8885	0.8916
SOLNet (Ours)	**0.9171**	**0.0078**	**0.9596**	**0.9577**	**0.8392**	**0.8513**	**0.9284**	**0.0111**	**0.9697**	**0.9689**	**0.8925**	**0.8946**
w/o $L^{BCE} + L_{F1}^M$	0.9097	0.0090	0.9476	0.9496	0.8281	0.8339	0.9266	0.0113	0.9661	0.9667	0.8823	0.8821
w/o $L^{IOU} + L_{F1}^M$	0.8919	0.0101	0.8633	0.9160	0.6713	0.7793	0.9204	0.0129	0.9326	0.9503	0.8045	0.8572
w/o L_{F1}^M	0.9169	0.0083	0.9541	0.9572	0.8154	0.8408	**0.9318**	**0.0110**	0.9676	0.9680	0.8778	0.8881
SOLNet (Ours)	**0.9171**	**0.0078**	**0.9596**	**0.9577**	**0.8392**	**0.8513**	0.9284	0.0111	**0.9697**	**0.9689**	**0.8925**	**0.8946**

Bold data represents the best results in the corresponding tag.

refined and closer to the ground truth compared to other methods.

2) Analysis of Network Structure and Loss Function: In the first part of Table IV, we analyze the impact of network structure. If the proposed LGA and EDE are not used to mine deep features in high-level feature maps, the overall accuracy of the network will drop significantly. Subsequently, after introducing LGA to enhance channel feature attention or using EDE to enhance spatial information attention, the accuracy of the network under six metrics is improved to varying degrees. However, relying solely on one module to focus on feature information in one direction is evidently insufficient. Therefore, only when LGA and EDE work together, the network could take into account both channel and spatial feature information, thereby further improving the accuracy of the network under all metrics. The results show that both LGA and EDE are favorable to improving the accuracy, which is also the key to high-precision decoding of SOLNet.

In the second part, we sequentially test the impact of different loss functions on the training results of SOLNet. As shown in Table IV, during the network training process, without using L^{IOU} or L^{BCE} to constrain the network, the training accuracy of the network is unsatisfactory. When L^{IOU} and L^{BCE} are used simultaneously, the resulting network model has high accuracy. On this basis, we additionally introduce a small perturbation loss term to further improve the training accuracy of the model. The results show that using the synergistic constraint of L^{IOU} and L^{BCE} benefits the model training. Especially, when L_{F1}^M is added, the overall performance of SOLNet is further improved.

E. Discussion

Here, we discuss the limitations of our method and our future work. Based on research analysis of the computational costs of existing methods, we have studied a GPU-friendly lightweight SOD-RSI model that significantly reduces the computational burden on the GPU. While our method has demonstrated SOTA inference speed on a desktop GPU compared to most CNN-based methods, we have paid less attention to the adaptability of edge computing devices with lightweight models. Therefore, in future work, we need to shift our research focus toward model quantization and deployment to address this aspect. In addition, similar to lightweight research trends in other fields, studying lightweight models that take into account both accuracy and speed is an ideal goal. However, experimental comparisons of existing works indicate that lightweight optimization for the SOD-RSI task requires further attention. Specifically, future efforts should focus on improving the evaluation of Lg in models.

V. CONCLUSION

In this article, we propose a novel speed-oriented lightweight SOD-RSI network, SOLNet, which has significant advantages in detection speed. Specifically, a DEAM is proposed to deconstruct–interact–recombine channel features and dynamically capture spatial information, which consists of an LGA module and a novel EDE module. Based on the view of "Deep decoding, once is enough," SOLNet only needs to mine high-level feature maps and perform simple cascade operations without paying additional attention to feature maps of other scales, greatly reducing the computational burden of the GPU. Especially, we also propose the concept of Lg, an evaluation metric that takes into account both speed and accuracy. It not only effectively reveals the under-gain issue of lightweight models but also provides theoretical support for the evaluation of subsequent lightweight works. Experimental results on the EORSSD and ORSSD datasets show that SOLNet achieves a leap in speed and is a SOTA lightweight SOD-RSI method.

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